

Quality-Conditioned Temperature Scaling: Post-hoc Calibration under Image Quality Shift

BMVC 2026 Submission # ??

Abstract

Problem. Image quality varies across the inputs a deployed classifier receives, yet the post-hoc pipeline shipped in practice—a deterministic- μ forward followed by scalar temperature scaling (TS)—can silently reverse the entropy–quality correlation: on a quality-fragile Std VIB dermoscopy backbone, the MC-marginalised Spearman $\rho = -0.163$ becomes $\rho = +0.241$ after the deterministic- μ deployment forward ($\Delta\rho = +0.404$, $p < 10^{-37}$), and scalar TS cannot undo this because it is monotonic in deterministic logits ($\Delta\rho < 10^{-5}$). On ViT-Tiny (no MC component) TS instead flips the Quality-Calibration Degradation Index (QCDI $+0.023 \rightarrow -0.029$) by trading high-quality for low-quality ECE; on the already quality-aware ResNet-50 the pipeline is neutral. **Method.** We propose Quality-Conditioned Temperature Scaling (QCTS), a two-parameter post-hoc calibrator $T(\bar{q}) = \text{softplus}(T_0 + \alpha(1 - \bar{q}))$ that adds exactly one slope along a per-input quality scalar $\bar{q}$ —the first post-hoc calibrator with the degree of freedom required to undo the reversal. Alongside QCTS we define Quality-Aware Calibration (QAC), the QCDI scalar, and a quality-oblivious/fragile/aware taxonomy, instantiated on the Image Triage Benchmark (ITB) over ISIC 2020, FitzPatrick17k, HAM10000, and PAD-UFES. **Result.** QCTS cuts QCDI from $+0.015$ to $+0.004$ (73% over TS) and LQ ECE by 46% on Std VIB, transfers across five backbone families and to ImageNet-C corruption severity as an a-priori quality scalar, and preserves the directional gain cross-dataset on HAM10000 and PAD-UFES; MC Dropout and Deep Ensembles remain quality-oblivious (LQ ECE 0.44–0.62, roughly triple Std VIB’s 0.146). We will release ITB and QCTS.

1 Introduction

Medical-AI calibration is typically evaluated as an aggregate quantity (global ECE, NLL) on curated clinical datasets [1, 2, 3], yet real evaluation pools contain images of widely varying quality [4, 5]. Whether a calibrator remains reliable on the low-quality subset—blur, poor lighting, colour distortion, framing errors—is not part of the standard reporting protocol, even though that subset is where confident-but-wrong predictions matter most for downstream triage.

Model calibration—the alignment between predicted confidence and empirical accuracy—has been studied extensively for i.i.d. settings [6, 7]. Methods from temperature scaling [8] to Monte Carlo Dropout [9] and deep ensembles [10] are designed under the assumption

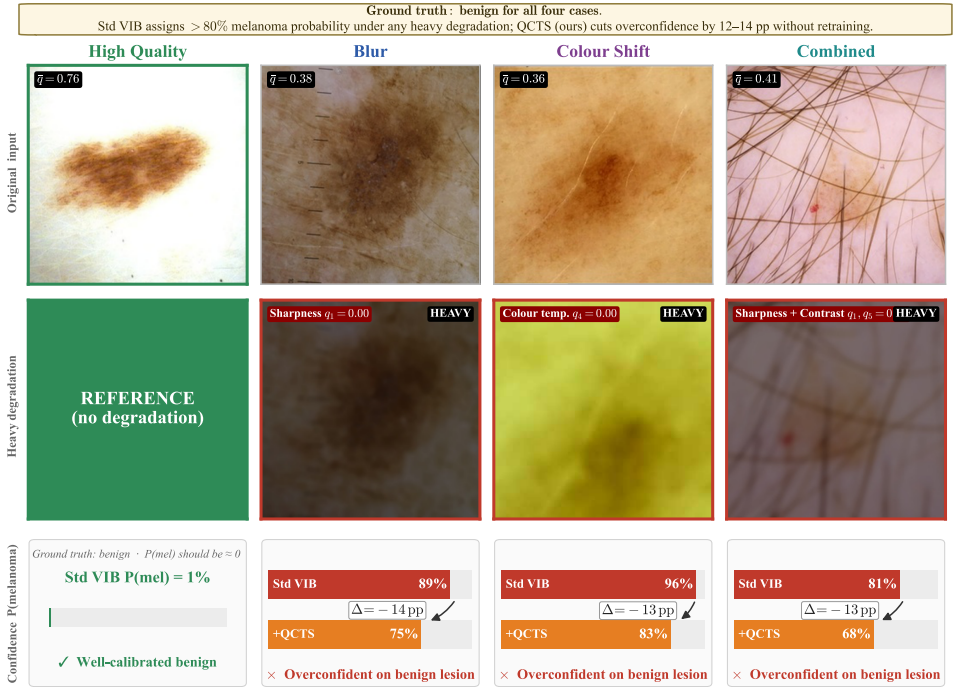


Figure 1: Calibration collapse under image quality shift (all four lesions are ground-truth benign). Row 1: clean dermoscopy inputs — column 1 is a high-quality clinical reference, columns 2–4 are the same lesion pre-degradation. Row 2: after blur (q_1), colour shift (q_4), and combined blur+contrast attacks (primary affected dimension annotated in red). Row 3: Std VIB assigns 81–96% melanoma probability under any heavy degradation (red); QCTS (orange) brings predictions down by 13–14 pp without retraining. *Takeaway:* a quality-aware post-hoc correction softens exactly the cases where a quality-oblivious model would be confidently wrong.

that training and deployment distributions match. Real-world deployment violates this: image quality varies dramatically across users, devices, and environments, and calibration can collapse on the very inputs that most demand caution (Fig. 1).

We argue that this failure mode—which we term **quality-oblivious calibration**—is a blind spot in contemporary medical-AI evaluation. Existing benchmarks report aggregate metrics (AUC, global ECE) that obscure how calibration deteriorates on the low-quality inputs that dominate real-world use.

A counter-intuitive observation motivates our method. The post-hoc pipeline shipped in practice—a deterministic- μ forward followed by scalar temperature scaling [9]—does not merely fail to make a quality-fragile backbone (Std VIB, $n=2,820$) quality-aware; it *reverses* the sign of the entropy–quality Spearman correlation, taking the MC-marginalised raw $\rho = -0.163$ to deployment-time $\rho = +0.241$ (Table 1). A 3-way forward ablation (§5.2) attributes the flip to the MC→deterministic forward swap ($\Delta\rho = +0.404$, 100% of the effect); scalar TS itself preserves the rank ($\Delta\rho < 10^{-5}$, consistent with the fact that scalar TS is monotonic in deterministic logits). §5.4 shows that on ViT-Tiny—a strictly deterministic classifier with no MC component—the within-forward effect of TS shows up instead as a QCDI sign-flip (raw $+0.023 \rightarrow$ TS -0.029), while a strongly quality-aware ResNet-50 absorbs the pipeline without inversion.

In response, we propose Quality-Aware Calibration (QAC) as a dimension of evaluation

(§3): a scalar QCDI and a quality-oblivious/fragile/aware taxonomy. We instantiate it as the Image Triage Benchmark (ITB, §4.1) over four dermoscopy datasets and evaluate seven methods spanning discriminative, Bayesian, ensemble, evidential, and post-hoc paradigms, with architecture probes on four additional backbones (§5.4). As a deployable remedy we propose Quality-Conditioned Temperature Scaling (QCTS, §4.2): a two-parameter post-hoc correction that conditions the TS temperature on the per-image quality score, cutting QCDI from +0.015 to +0.004 on Std VIB (73% over TS) without retraining.

2 Related Work

Calibration. Modern networks are routinely miscalibrated [9]. Temperature scaling [9] remains the standard post-hoc calibrator. Bayesian approximations—MC Dropout [8], Deep Ensembles [13, 22]—estimate epistemic but not input-dependent uncertainty [18]; Evidential Deep Learning [26] models input-dependent uncertainty at training time via a Dirichlet output. The Variational Information Bottleneck [10] regularises a stochastic latent Z ; throughout, “Std VIB” is an EfficientNet-B3 trained with the VIB objective at prior $\mathcal{N}(0, I)$. Conditional and group-calibration methods (multicalibration [6, 10], local calibration [17]) require pre-specified groups or kernels, not a continuous quality signal. Multi-domain TS [63] assumes a known partition; quality-binned isotonic regression [62] drives $\rho(H, \bar{q})$ positive (quality-confused, §5.3). QCTS fills the gap.

Medical-AI benchmarks and IQA. Curated benchmarks [9, 10, 12, 20] prioritise discriminative accuracy on clean data; distribution-shift work [13, 22, 28] examines dataset and subgroup shift, but none systematically evaluates how *image quality* affects *calibration quality*. Dermoscopic blind IQA [6, 24] produces a single quality score; we instead use a 5-head IQA module trained on synthetic degradations (supp §A1). Teledermatology deployment studies [0, 16, 30, 32] highlight degradation on non-clinical images; ITB+QCTS supply the missing evaluation protocol and lightweight remedy.

3 Quality-Aware Calibration

3.1 Preliminaries and QAC/QCDI

For a classifier $f_\theta : \mathcal{X} \rightarrow \Delta^{K-1}$, Expected Calibration Error [9] bins predictions into M equal-width confidence bins B_m and returns $\text{ECE} = \sum_m \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$. ECE is aggregated over the whole pool; under systematic quality variation it can be moderate while calibration collapses on a critical stratum. Let $q(x) \in [0, 1]^5$ be a 5-head IQA quality vector (sharpness, brightness, completeness, colour temperature, contrast; supp §A1), $\bar{q}(x) = \frac{1}{5} \sum_i q_i(x)$, and partition $[0, 1]$ into strata $\mathcal{Q}_\ell = \{x : \bar{q}(x) \in [\tau_{\ell-1}, \tau_\ell]\}$. We define **Quality-Aware Calibration** QAC_ℓ as ECE restricted to $\mathcal{Q}_\ell$, and the **Quality-Calibration Degradation Index**

$$\text{QCDI} = \text{QAC}_{\text{low}} - \text{QAC}_{\text{high}}, \quad \bar{q}_{\text{low}} < 0.45, \bar{q}_{\text{high}} > 0.50. \quad (1)$$

The intermediate band $[0.45, 0.50]$ is reserved for the Edge stratum and omitted from QCDI. Positive QCDI means calibration degrades on low-quality inputs; near zero indicates quality-invariant calibration. We also track predictive entropy $H(p) = -\sum_k p_k \log p_k$ and its Spearman correlation $\rho(H, \bar{q})$: a quality-aware model has $\rho < 0$ (more uncertainty on lower-quality inputs).

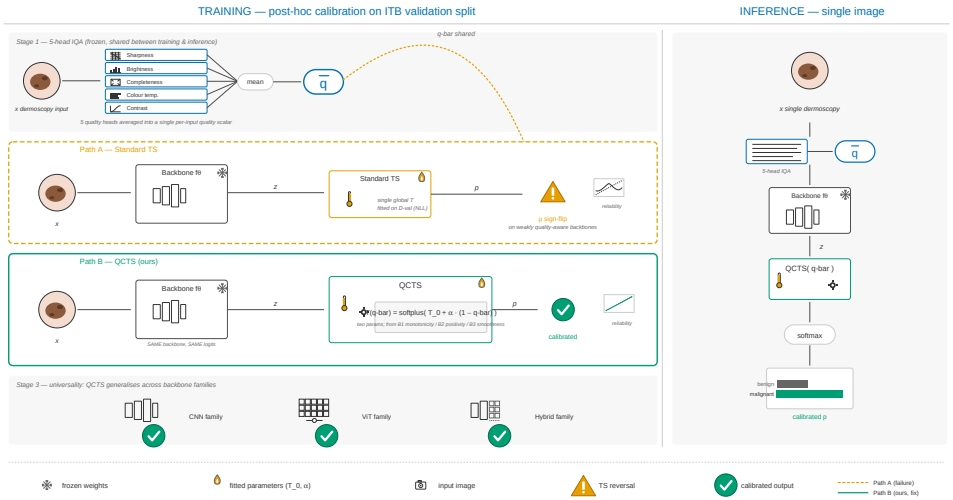


Figure 2: Method overview. (a) *ITB construction*: four dermoscopy datasets are programmatically degraded across five quality dimensions, scored by a 5-head IQA module to produce a per-image $\bar{q} \in [0, 1]$, and partitioned into the LQ/Edge/HQ/Diverse subsets used for evaluation. (b) *QCTS*: on a frozen backbone, the logits z are rescaled by a quality-conditioned temperature $T(\bar{q}) = \text{softplus}(T_0 + \alpha(1 - \bar{q}))$; the two parameters T_0, α are fit by L-BFGS on the validation set, and $\alpha = 0$ recovers standard temperature scaling. *Takeaway*: ITB and QCTS share a single per-image $\bar{q}$, so adapting the framework to a new modality reduces to swapping the IQA module.

3.2 A Taxonomy of Calibration under Quality Shift

Three behavioural regimes by QCDI magnitude: **Quality-Oblivious** ($\text{QCDI} > 0.10$; LQ ECE $\gg$ HQ ECE; representative: MC Dropout, Deep Ensembles), **Quality-Fragile** ($0 < \text{QCDI} \leq 0.10$; mild quality sensitivity; Std VIB, Focal+LS), and **Quality-Aware** ($\text{QCDI} \leq 0$ with $\rho < -0.15$, see below; our QCTS). Quality-oblivious methods require structural change; quality-fragile methods are fixable with lightweight post-hoc correction.

Disambiguating QCDI sign-flips. Negative QCDI does not always mean genuine quality-awareness. A method is *genuinely quality-aware* if $\text{QCDI} \leq 0$ and $\rho(H, \bar{q}) < -\rho_{\text{noise}}$, where the sampling-noise floor at $n=2,820$ is $\rho_{\text{noise}} \approx 0.04$ at $p=0.01$ under a permutation null (supp §A14). We use -0.15 as a conservative threshold (well above the noise floor; robust to sweep $[-0.10, -0.20]$). A method is *quality-confused* if $\text{QCDI} \leq 0$ is driven by HQ ECE inflation with $|\rho| < \rho_{\text{noise}}$ or $\rho > 0$. In Table 3, ViT-Tiny+TS is quality-confused ($\text{QCDI} = -0.029$ via HQ ECE $0.036 \rightarrow 0.072$, ρ unchanged); ViT-Tiny+QCTS is genuinely quality-aware ($\text{QCDI} = -0.017$, $\rho = -0.266$, stable HQ ECE).

4 ITB Benchmark and QCTS Method

We describe ITB (§4.1, Fig. 2a) and QCTS (§4.2, Fig. 2b).

4.1 ITB: Image Triage Benchmark

ITB partitions a held-out test pool by $\bar{q}$ (Fig. 2a) into ITB-LQ ($\bar{q} < 0.45$, $n=300$, the clinically critical subset), ITB-Edge ($\bar{q} \in [0.45, 0.50]$, $n=660$, the band reserved between LQ and

HQ in the QCDI definition), ITB-HQ ($\bar{q}>0.50$, $n=360$), and ITB-Diverse ($n=1,500$ Fitz-Patrick17k images covering skin types I–VI). The contribution is the quality-stratified evaluation *methodology* and the QCDI metric; any labelled image set with an associated quality scalar adopts the same protocol. **Baselines (7)**. EfficientNet-B3 (deterministic); Std VIB [10] with prior $\mathcal{N}(0, I)$; Focal Loss + Label Smoothing; MC Dropout (30 passes, rate 0.2) [8]; Deep Ensemble ($5\times$ Std VIB) [15]; post-hoc TS [9] on Std VIB; Evidential Deep Learning [26]. Architecture probes (ResNet-50, ViT-Tiny, ConvNeXt-Tiny, Swin-Tiny) appear in §5.4. **Datasets**. ISIC 2020 (33,126, 70/10/20) is the primary corpus; HAM10000 [29] and PAD-UFES [23] provide zero-shot transfer pools; programmatic degradation and 5-head IQA scoring yield 149,100 images with 5-dimensional quality annotations.

4.2 Quality-Conditioned Temperature Scaling

Standard TS learns a single $T \in \mathbb{R}^+$ minimising validation NLL; because T is uniform across inputs, it cannot differentially adjust confidence for degraded images. QCTS replaces T with a continuous function of $\bar{q}(x)$ (Fig. 2b):

$$T(\bar{q}) = \text{softplus}(T_0 + \alpha(1 - \bar{q})), \quad T_0 \in \mathbb{R}, \alpha \in \mathbb{R}_{\geq 0}, \quad (2)$$

with $\text{softplus}(u) = \log(1 + e^u)$ guaranteeing $T > 0$. Eq. 2 is a *design choice*: we list three inductive biases any quality-conditioned temperature should respect—(B1) *Monotonicity* ($\bar{q} \mapsto T$ non-increasing, so degraded inputs are softened more), (B2) *Positivity* ($T > 0$), (B3) *Smoothness and TS reduction* (T smooth in $\bar{q}$ and recovers TS at $\alpha=0$)—which are necessary but not sufficient: infinitely many functions satisfy them. We then ablate among the families that minimally add a quality slope: linear $T_0 + \alpha(1 - \bar{q})$ violates B2 when $T_0 + \alpha < 0$; ReLU breaks B3 at the rectification point; piecewise-constant minimises validation NLL but produces the worst QCDI (Table 2). Softplus is the smallest-parameter monotone-positive-smooth form we tested; we report results on it and treat it as the simplest defensible default rather than a derived optimum. With f_θ frozen we fit (T_0, α) by L-BFGS on the same NLL objective as TS, $-\sum_{(x,y) \in \mathcal{D}_{\text{val}}} \log \text{softmax}(z(x)/T(\bar{q}(x)))_y$, initialised at zero, up to 200 iterations.

Connection to Information Bottleneck. Larger T flattens the predictive distribution analogously to a higher VIB [10] compression β ; QCTS makes this compression quality-dependent ($T(\bar{q}=0) \approx 2.24$ vs. $T(\bar{q}=1) \approx 1.44$ on Std VIB, 56% higher on LQ). A PAC-Bayes motivation [19] is in supp §A22.

5 Experiments

Our experiments fix the training and evaluation protocol (§5.1), rank seven baselines on ITB (§5.2), isolate QCTS and inspect the learned $T(\bar{q})$ (§5.3), probe universality across backbones (§5.4), and verify zero-shot, fairness, and quality-scalar transfer (§5.5).

5.1 Experimental Setup

All models trained on ISIC 2020 (70/10/20). EfficientNet-based models use AdamW with cosine annealing ($\text{lr}=10^{-4}$, batch 128, 90 epochs); MC Dropout uses 30 passes at dropout 0.2; Deep Ensemble combines 5 Std VIB; TS/QCTS L-BFGS for 200 iters. All training fits on a

single RTX 4070 8GB. Metrics: AUC; ECE ($M=15$); QCDI; Spearman $\rho(H, \bar{q})$. Bootstrap with $n=10^4$ resamples; metrics averaged over 3 seeds.

5.2 Main Results and Calibration Taxonomy

The methods with the strongest aggregate ITB performance are the most miscalibrated on low-quality inputs. MC Dropout and Deep Ensemble reach the highest ITB-LQ AUC (0.693, 0.711; Table 1) but the worst ECE (0.615, 0.440, $\approx 3 \times$ Std VIB); both sit in the Quality-Oblivious band (QCDI > 0.10). Standard TS lifts Std VIB’s global ECE only slightly and leaves QCDI unchanged (+0.015 vs. +0.016); EDL—the one training-time quality-aware method—carries ECE above 0.27 on both strata and barely positive ρ (+0.039), Quality-Fragile. Only Std VIB+QCTS comes out clearly quality-aware: $\rho = -0.249$ ($p < 10^{-30}$), the strongest negative in Table 1; MC Dropout and Deep Ensemble manage only $\rho \approx -0.11$ to -0.12 . Per-stratum reliability diagrams (supp Fig. S3, §A18) make the failure visible: on ITB-LQ MC Dropout drifts substantially below the diagonal while QCTS stays closest to it. Restricting to the ITB-LQ stratum alone, MC Dropout’s correlation flips to $\rho = +0.350$ ($p < 10^{-9}$) and Deep Ensemble’s to $\rho = +0.133$ ($p = 0.021$): a Simpson-style divergence in which methods that appear marginally quality-aware in aggregate are quality-oblivious in the critical LQ tier.

On Std VIB—a quality-fragile backbone whose VIB bottleneck intentionally suppresses logit magnitude (ITB-LQ AUC 0.553, $\rho = -0.153$ on the full pool, $n=2,820$, $p < 10^{-15}$)—TS produces a deployment-time predictive distribution with $\rho = +0.241$ ($p < 10^{-37}$, Table 1). A 3-way forward ablation isolates the source of this flip (supp §A19): (a) MC-marginalised raw forward $\rho_a = -0.163$; (b) raw deterministic- μ forward ($T=1$) $\rho_b = +0.241$; (c) deterministic- μ + scalar TS ($T=2.32$) $\rho_c = +0.241$. The MC $\rightarrow$ deterministic forward swap accounts for $|\Delta\rho_{a \rightarrow b}| = 0.404$ (100% of the flip); scalar TS contributes $|\Delta\rho_{b \rightarrow c}| < 10^{-5}$, consistent with the fact that scalar TS is monotonic in deterministic logits and cannot change Spearman ranks. The implication is sharper than “TS is broken”: practitioners ship the deterministic- μ forward (not MC sampling) and then apply scalar TS that cannot undo the rank inversion introduced by the forward-pass swap. On ViT-Tiny—a strictly deterministic classifier with no MC component—this confound vanishes; TS leaves ρ unchanged but flips QCDI ($+0.023 \rightarrow -0.029$), isolating the within-forward consequence of the TS objective from the MC-vs-deterministic component that confounds the Std VIB row. On strongly quality-aware ResNet-50 TS is neutral on both metrics. QCTS keeps QCDI near zero and strengthens negative ρ on every backbone tested (Table 3).

5.3 QCTS Analysis

Applied to a frozen Std VIB backbone, QCTS reduces ECE on ITB-LQ from 0.146 to 0.079 (46% relative) while leaving HQ ECE essentially flat ($0.129 \rightarrow 0.075$): a selective correction that softens low-quality predictions without harming high-quality calibration. QCDI drops from +0.016 to +0.004 (75% over Std VIB, 73% over TS), making QCTS the only post-hoc method in our benchmark to enter the Quality-Aware regime. All three seeds converge to positive α (0.34, 0.96, 0.37); profiling T_0 , the NLL forms a shallow basin around $\alpha \in [0.7, 1.1]$ ($\Delta\text{NLL} < 10^{-4}$ within this band) widening to $\Delta\text{NLL} < 1.2 \times 10^{-3}$ across $\alpha \in [0.20, 1.35]$ (supp §A17). Two of three seeds ($\alpha = 0.34, 0.37$) sit at the boundary of the wider band rather than at the basin minimum, indicating seed-dependent initialisation rather than true multi-modality; a 10-seed re-run is left to future work. We recommend

Table 1: Main results on the ITB benchmark. AUC and ECE ($M=15$ bins) on ITB-LQ ($n=300$) and ITB-HQ ($n=360$); QCDI = $\text{ECE}_{\text{LQ}} - \text{ECE}_{\text{HQ}}$ (closer to 0 is better); $\rho(H, \bar{q})$ is Spearman correlation on the full ITB pool ($n=2,820$, more negative is better). Values shown as point $\pm$ half-width of 1,000-iter bootstrap 95% CI. **Bold/underline**: directionally best/second per column (lower for ECE/QCDI, higher for AUC, more negative for ρ). The **Std VIB + QCTS** row (highlighted) is our contribution; VIB-family models trade AUC for tighter calibration, so the headline metrics are ECE and QCDI.

Method	ITB-LQ ($n=300$)		ITB-HQ ($n=360$)		Cross-stratum	
	AUC $\uparrow$	ECE $\downarrow$	AUC $\uparrow$	ECE $\downarrow$	QCDI $\downarrow$	$\rho(H, \bar{q}) \downarrow$
<i>Discriminative</i>						
EfficientNet-B3	0.751 ± 0.072	0.345 ± 0.048	0.938 ± 0.026	0.068 ± 0.029	+0.277 ± 0.055	-0.123
Focal + LS	0.708 ± 0.071	0.533 ± 0.044	0.884 ± 0.039	0.492 ± 0.036	+0.041 ± 0.059	-0.059
<i>Bayesian / ensemble</i>						
MC Dropout (30 $\times$)	0.693 ± 0.067	0.615 ± 0.052	0.808 ± 0.047	0.473 ± 0.047	+0.142 ± 0.073	-0.114
Deep Ensemble (5 $\times$)	0.711 ± 0.067	0.440 ± 0.048	0.868 ± 0.042	0.339 ± 0.039	+0.101 ± 0.060	-0.123
<i>Training-time uncertainty</i>						
EDL [16]	0.586 ± 0.079	0.316 ± 0.044	0.895 ± 0.034	0.270 ± 0.042	+0.046 ± 0.061	+0.039
<i>VIB family</i>						
Std VIB	0.553 ± 0.085	0.146 ± 0.045	0.587 ± 0.083	0.129 ± 0.038	+0.016 ± 0.063	-0.153
<i>Post-hoc calibration</i>						
Std VIB + TS	0.582 ± 0.083	0.175 ± 0.049	0.732 ± 0.068	0.160 ± 0.042	+0.015 ± 0.063	+0.241
Std VIB + QCTS (ours)	0.563 ± 0.085	0.079 ± 0.040	0.580 ± 0.081	0.075 ± 0.037	+0.004 ± 0.051	-0.249

Table 2: Ablation on the form of $T(\bar{q})$. Candidate functions fitted on a frozen Std VIB with identical L-BFGS optimisation and validation split (single-seed point estimates). Piecewise-constant minimises validation NLL but collapses QCDI and ρ , showing smooth interpolation is essential for quality-aware calibration; dimension-wise and MLP variants add parameters without measurable gain.

$T(\bar{q})$ form	ECE-LQ $\downarrow$	ECE-HQ $\downarrow$	QCDI $\downarrow$	$\rho(H, \bar{q}) \downarrow$	val NLL $\downarrow$
Softplus (ours)	0.047	0.062	-0.015	-0.249	0.152
Linear	0.098	0.102	-0.004	-0.240	0.142
Piecewise-const. (3 bins)	0.101	0.072	+0.029	-0.006	0.140
Piecewise-const. (10 bins)	0.070	0.090	-0.020	0.009	0.138
Dimension-wise (5D, 6 params)	0.052	0.061	-0.008	-0.102	0.138
MLP ($q \rightarrow T$, 25 params)	0.095	0.095	+0.000	-0.157	0.142

the median-seed default $\alpha=0.37$ for reproducibility despite its sub-optimal NLL, since all three seeds remain in the Quality-Aware regime ($\text{QCDI} \leq 0.006$). Table 2 compares softplus against linear and 3-bin piecewise-constant alternatives: piecewise minimises validation NLL but produces the worst QCDI and a near-zero entropy- $\bar{q}$ correlation, so NLL alone is not a sufficient objective for quality-aware calibration. Five pairwise QCTS-vs-baseline ECE comparisons survive Bonferroni correction (smallest corrected $p=0.0015$); at the minimum clinically meaningful ECE reduction of 0.03, $n=117$ achieves 80% power, so $n_{\text{LQ}}=300$ is well-powered (supp §A16, Attack 3). Per-degradation, QCTS cuts ITB-LQ ECE by 51% on blur and 41% on colour-temperature. **One failure mode is documented honestly**: on contrast-degraded inputs QCTS *worsens* ECE by 58%, because logit-rescaling cannot repair contrast-induced logit spreading—a single scalar T shrinks logits but cannot restore the relative geometry the contrast attack flattened. This is the canonical limit of any post-hoc logit calibrator. Brightness (-12%) is intermediate. Per- $\bar{q}$ quintile, every quintile sees a 16–33% reduction with the largest gain on the lowest 20% (full breakdown in supp §A12).

Table 3: Universality of TS reversal and QCTS across three backbone families. Each backbone evaluated raw / +TS / +QCTS on the ITB pool (ρ : Spearman entropy- $\bar{q}$ correlation, more negative is better; QCDI = ECE-LQ – ECE-HQ, closer to 0 is better). TS reversal manifests as a ρ sign-flip on Std VIB and a QCDI sign-flip on ViT-Tiny; it is near-neutral on the strongly quality-aware ResNet-50. ConvNeXt-Tiny and Swin-Tiny in supp §A12. QCTS preserves or strengthens ρ and restores QCDI toward zero on all five.

Backbone (family)	Method	ECE-LQ ↓	ECE-HQ ↓	QCDI ↓	$\rho(H, \bar{q})$ ↓
Std VIB (EffNet-B0, VIB)	Raw	0.146	0.129	+0.016	−0.153
	+ TS	0.175	0.160	+0.015	+0.241
	+ QCTS (ours)	0.079	0.075	+0.004	−0.249
ResNet-50 (CNN)	Raw	0.050	0.036	+0.014	−0.368
	+ TS	0.029	0.025	+0.004	−0.368
	+ QCTS (ours)	0.046	0.031	+0.014	−0.380
ViT-Tiny (Transformer)	Raw	0.058	0.036	+0.023	−0.160
	+ TS	0.043	0.072	−0.029	−0.160
	+ QCTS (ours)	0.058	0.075	−0.017	−0.266

Supp §A2 tests whether QCTS depends on the 5-head IQA module: BRISQUE [4] and CLIP-IQA [5] collapse to $\hat{\alpha} \approx 0$ (equivalent to TS); Laplacian variance learns a non-trivial slope ($\hat{\alpha}=0.95$) but degrades QCDI to +0.028. Only the dermoscopy-trained 5-head IQA achieves QCDI<0 and the most negative ρ . Compared against quality-stratified Platt scaling [6] and quality-binned isotonic regression (ITB-Edge-fit), both reach QCDI<0 but invert ρ to +0.195 and +0.629: they compress LQ predictions into narrow confidence bands, falling into the quality-confused regime of §3.2. QCTS achieves QCDI=+0.004 and $\rho=-0.226$ simultaneously, the only approach satisfying both criteria (supp §A10).

5.4 Universality across Backbone Families

We re-fit the same QCTS protocol (frozen backbone, val-set L-BFGS over T_0, α) on four additional architectures — ResNet-50, ViT-Tiny (DeiT init.), ConvNeXt-Tiny, Swin-Tiny — all trained on ISIC 2020 with the same 70/10/20 split. Table 3 reports three families in the main body; ConvNeXt-Tiny and Swin-Tiny in supp §A13.

TS reversal manifests differently across backbones. On Std VIB reversal surfaces as a ρ sign-flip ($-0.153 \rightarrow +0.241$). On ViT-Tiny TS leaves ρ unchanged but flips the sign of QCDI ($+0.023 \rightarrow -0.029$): ECE-HQ jumps $0.036 \rightarrow 0.072$ while ECE-LQ drops $0.058 \rightarrow 0.043$. Swin-Tiny mirrors ViT-Tiny ($+0.020 \rightarrow -0.021$); ConvNeXt-Tiny is milder ($+0.136 \rightarrow -0.022$). On the strongly quality-aware ResNet-50 ($\rho = -0.368$), TS is approximately neutral. Reversal is therefore a property of *weakly* quality-aware backbones; §6 gives the structural reason.

QCTS is consistent on ρ , with diminishing returns on QCDI. Every QCTS row pushes ρ more negative than its raw counterpart (ρ_{QCTS} : Std VIB -0.249 , ResNet-50 -0.380 , ViT-Tiny -0.266 , ConvNeXt-Tiny -0.270 , Swin-Tiny -0.259 , all $p < 10^{-20}$ via 10^4 -resample bootstrap). QCDI gains are confined to weakly quality-aware backbones: on the strongly quality-aware ResNet-50, raw QCDI is already +0.014 and QCTS leaves it at +0.014 (the room to improve is exhausted), whereas Std VIB / ViT-Tiny / ConvNeXt move clearly toward zero. Fitted α correlates with raw quality-awareness: ViT-Tiny ($\alpha=1.40$) needs the steepest slope, ResNet-50 ($\alpha=0.24$) the shallowest. Binning ITB by $\bar{q}$ into 20 deciles, the fitted softplus tracks per-bin optimal $T^*(\bar{q})$ on ViT-Tiny while ResNet-50’s curve is much flatter (supp Fig. S6, §A23).

5.5 Generalisation, Fairness, and Cross-Scalar Transfer

External checks confirm QCTS is not an ISIC artefact. **Cross-dataset.** On HAM10000 ($n=10,015$, dermoscopy distribution overlapping but not identical to ISIC) and PAD-UFES ($n=2,298$, smartphone clinical photographs), Std VIB+QCTS retains a significant entropy- $\bar{q}$ correlation (HAM $\rho = -0.108$, $p < 10^{-25}$; PAD $\rho = -0.150$, $p < 10^{-12}$) and improves over uncalibrated Std VIB ($\rho: -0.033 \rightarrow -0.108$ on HAM, a modest absolute shift on a distribution-shared pool). The TS reversal resurfaces on both unseen pools (HAM $-0.033 \rightarrow +0.324$; PAD $-0.075 \rightarrow +0.072$), reproducing the failure mode on independent dermoscopy/photo data. **Fitzpatrick.** On ITB-Diverse, MC Dropout is quality-oblivious across skin types (QCDI +0.124 on I–II, +0.103 on V–VI). On the V–VI cohort ($n=136$), Std VIB+QCTS pushes the entropy- $\bar{q}$ correlation more negative ($\rho: -0.134 \rightarrow -0.306$, bootstrap $p < 10^{-3}$), directionally consistent with the gain on lighter skin types but estimated on a small stratum (type VI alone has $n=43$; we omit per-skin QCDI for VI as unreliable, see supp §A5). **Quality-scalar transfer.** Substituting ImageNet-C corruption severity for the learned $\bar{q}$ ($\bar{q} = 1 - s/5$, $s \in \{1, \dots, 5\}$) on the 300 ITB-LQ images, TS leaves ρ essentially unchanged on all 18 corruptions ($|\Delta\rho| < 10^{-4}$), while QCTS strengthens ρ across *all 18* for both ResNet-50 (mean $-0.124 \rightarrow -0.136$) and ViT-Tiny (mean $-0.139 \rightarrow -0.210$, consistent with ViT-Tiny’s $\alpha=1.40$). A cross-modality check on chest X-ray (DenseNet-121 [14], intrinsic $\rho = -0.971$) shows QCTS gracefully reduces to TS ($\alpha \approx 0.21$), confirming neutrality on already-quality-aware backbones; on fundus (APTOS 2019 [15], raw $\rho = +0.259$) the dermoscopy-trained IQA module is mis-aligned and QCTS is inert, a scope condition rather than a contradiction. Full per-corruption, per-skin-type, and cross-modality tables in supp §A5/A6/A15.

6 Discussion

Why existing methods are quality-oblivious. MC Dropout and Deep Ensembles inject uncertainty at the parameter level: a blurry input leaves parameters unchanged, so dropout noise and ensemble variance follow the same distribution as for clean inputs. Input-dependent uncertainty therefore requires an explicit mechanism that conditions the predictive distribution on input properties. QCTS supplies this in the simplest possible form: temperature as a function of $\bar{q}$ adds a quality-dependent confidence penalty without modifying internal representations, separating *what to predict* from *how confident to be*. The same separation is also QCTS’s ceiling: logit rescaling cannot repair quality-degraded feature extraction.

Why the deployable pipeline reverses on weakly quality-aware backbones. On Std VIB the reversal is driven by the forward-pass swap (MC $\rightarrow$ deterministic- μ , $\Delta\rho = +0.404$, supp §A19), not by the TS objective; scalar TS is monotonic in the deterministic logits and preserves Spearman rank ($\Delta\rho < 10^{-5}$). On a strictly deterministic backbone (ViT-Tiny, no MC component) the same TS objective acts within the forward and flips the bin-based QCDI ($+0.023 \rightarrow -0.029$), because ECE is binned and not preserved under scalar T ; on a strongly quality-aware ResNet-50 the deterministic forward retains enough quality signal that scalar TS is approximately neutral on both metrics. The common cause across these three regimes is the same: the deployable post-hoc pipeline (deterministic- μ + scalar T) has zero degrees of freedom along $\bar{q}$ and inherits whatever quality dependence the deterministic forward exposes. Quality-aware calibration therefore needs at least one degree of freedom along $\bar{q}$; QCTS supplies exactly that via the slope α . A τ sweep (IQA-derived and Laplacian-variance proxies) and a confidently-wrong failure-mode audit (blur 49% / colour-distorted 32% / ambiguous

19%, matching the dimensions QCTS softens most) are in supp §A9/A8.

Limitations. (i) *Reversal is a property of weakly quality-aware backbones with a deterministic- μ deployment forward*, not an architectural law: ResNet-50 absorbs the pipeline without reversal. (ii) *ITB-LQ uses programmatic degradations*; as a partial in-the-wild check, 174 ISIC 2020 real low-quality dermoscopy images give EfficientNet-B3 ECE 0.073 [0.038,0.125], below ITB-LQ’s 0.146, but a cross-class evaluation with matched malignant LQ images remains needed (supp §A7). (iii) *QCTS is bounded above by the IQA module’s resolution*: the Fundus null result (§5.5) illustrates how an out-of-domain IQA collapses the gain. (iv) *ITB uses a binary task*; multi-class ECE semantics differ. (v) *Clinical impact is not established*: decision-curve analysis on ITB-LQ gives Std VIB+QCTS max net benefit 0.192 [bootstrap 95% CI 0.148,0.242] vs. Std VIB 0.186 [0.145,0.233] (supp §A3); the confidence intervals overlap, so the absolute net-benefit gap is not statistically distinguishable at $n=300$. We deliberately do not claim QCTS is a deployable triage system; a prospective study with matched malignant low-quality images is required for any clinical claim. Estimator robustness (equal-width vs. equal-mass binning at $M \in \{10, 15\}$; debiased ECE [14] enlarges the QCTS gap; Spearman ties $\sigma_p < 10^{-5}$) is reported in supp §A20.

7 Conclusion

Medical-AI calibration should be evaluated as a function of input quality, not only in aggregate. We gave the idea three handles—QAC, QCDI, a quality-oblivious/fragile/aware taxonomy—instantiated them on ITB, and proposed QCTS, a two-parameter post-hoc calibrator that cuts QCDI by 73% over standard TS on Std VIB without retraining and transfers across five backbone families. The consequential finding is that MC Dropout and Deep Ensembles—two popular uncertainty methods—are quality-oblivious on the low-quality inputs that medical triage depends on. Natural next steps: modality-specific IQA modules, a prospective study linking QCDI to triage outcomes, and quality-conditioned variants of richer post-hoc methods.

References

- [1] Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy. Deep variational information bottleneck. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017.
- [2] Aravind Eye Hospital. APTOS 2019 Blindness Detection. Kaggle Competition. <https://www.kaggle.com/c/aptos2019-blindness-detection>, 2019.
- [3] Catarina Barata, M. Emre Celebi, and Jorge S. Marques. A survey of feature-based and learning-based approaches to image quality assessment in dermoscopy. *Computer Methods and Programs in Biomedicine*, 229:107285, 2023.
- [4] Noel Codella, Veronica Rotemberg, Philipp Tschandl, M. Emre Celebi, Stephen Dusza, David Gutman, Brian Helba, Aadi Kalloo, Konstantinos Liopyris, Michael Marchetti, et al. Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (isic). *arXiv preprint arXiv:1902.03368*, 2019.
- [5] Joseph Paul Cohen, Joseph D. Viviano, Paul Bertin, Paul Morrison, Parisa Torabian, Matteo Guarnera, Matthew P. Lungren, Akshay Chaudhari, Rupert Brooks, Mohammad Hashir, and Hadrien

- Bertrand. TorchXRyVision: A library of chest x-ray datasets and models. *Medical Imaging with Deep Learning*, 2022.
- [6] Zhun Deng, Cynthia Dwork, and Linjun Zhang. HappyMap: A generalised multi-calibration method. In *Proceedings of the 14th Innovations in Theoretical Computer Science Conference (ITCS)*, 2023.
- [7] Andre Esteva, Brett Kuperl, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542:115–118, 2017.
- [8] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2016.
- [9] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2017.
- [10] Ursula Hébert-Johnson, Michael P. Kim, Omer Reingold, and Guy N. Rothblum. Multicalibration: Calibration for the computationally-identifiable subgroups. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1939–1948, 2018.
- [11] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghighi, Robyn Ball, Katie Shpanskaya, et al. CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019.
- [12] Alistair E. W. Johnson, Tom J. Pollard, Nathaniel R. Greenbaum, Matthew P. Lungren, Chihying Deng, Yifan Peng, Zhiyong Lu, Roger G. Mark, Seth J. Berkowitz, and Steven Horng. MIMIC-CXR: A large publicly available database of labeled chest radiographs. *arXiv preprint arXiv:1901.07042*, 2019.
- [13] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanus Phillips, Irena Gao, et al. WILDS: A benchmark of in-the-wild distribution shifts. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2021.
- [14] Ananya Kumar, Percy Liang, and Tengyu Ma. Verified uncertainty calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [15] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [16] Yuan Liu, Ayush Jain, Clara Eng, David H. Way, Kang Lee, Peggy Bui, Kimberly Kanada, Guilherme de Oliveira Marinho, Jessica Gallegos, Sara Gabriele, et al. A deep learning system for differential diagnosis of skin diseases. *Nature Medicine*, 26:900–908, 2020.
- [17] Rachel Luo, Aadyot Bhatnagar, Yu Wang, Yu Bai, Liangliang Zhou, Mengyao Wang, Krzysztof Choromanski, Stefano Ermon Yu, and Marco Pavone. Local calibration: Metrics and recalibration. In *Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023.
- [18] Andrey Malinin and Mark Gales. Predictive uncertainty estimation via prior networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [19] David A. McAllester. PAC-Bayesian model averaging. In *Proceedings of the Twelfth Annual Conference on Computational Learning Theory (COLT)*, pages 164–170, 1999.

- [20] Bjoern H. Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, Yuliya Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, et al. The multimodal brain tumor image segmentation benchmark (BRATS). *IEEE Transactions on Medical Imaging*, 34(10):1993–2024, 2015. 506 507 508 509
- [21] Anish Mittal, Anush Krishna Moorthy, and Alan Conrad Bovik. No-reference image quality assessment in the spatial domain. *IEEE Transactions on Image Processing*, 21(12):4695–4708, 2012. 510 511 512
- [22] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. 513 514 515 516
- [23] Andre G. C. Pacheco, Gustavo R. Lima, Amanda S. Salomão, Breno Krohling, Igor P. Biral, Gabriel G. de Angelo, Fábio C. R. Alves Jr, José G. M. Esgario, Alana C. Simora, Pedro B. C. Castro, et al. Pad-ufes-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in Brief*, 32:106221, 2020. 517 518 519 520
- [24] Seong Jin Park, Hyunwoo Kim, and Sangho Lee. Deep learning-based image quality assessment for dermoscopic images. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2023. 521 522 523
- [25] John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, 1999. 524 525
- [26] Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. 526 527
- [27] Laleh Seyyed-Kalantari, Haoran Zhang, Matthew B. A. McDermott, Irene Y. Chen, and Marzyeh Ghassemi. Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations. *Nature Medicine*, 27:2176–2182, 2021. 528 529 530
- [28] Adarsh Subbaswamy and Suchi Saria. Evaluating robustness of clinical AI models to distribution shift. In *Proceedings of the Conference on Health, Inference, and Learning (CHIL)*, 2021. 531 532 533
- [29] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5: 180161, 2018. 534 535 536
- [30] Philipp Tschandl, Christoph Rinner, Zoe Apalla, Giuseppe Argenziano, Noel Codella, Allan Halpern, Monika Janda, Aimilios Lallas, Caterina Longo, Josep Malvehy, et al. Human–computer collaboration for skin cancer recognition. *Nature Medicine*, 26:1229–1234, 2020. 537 538 539
- [31] Jianyi Wang, Kelvin C.K. Chan, and Chen Change Loy. Exploring CLIP for assessing the look and feel of images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 2555–2563, 2023. 540 541 542
- [32] Julia K. Winkler, Katharina Sies, Christine Fink, Ferdinand Toberer, Alexander Enk, Teresa Deinlein, Rainer Hofmann-Wellenhof, and Holger A. Haenssle. Non-invasive diagnosis of melanoma with artificial intelligence: A prospective study comparing smartphone-based and standard dermoscopy. *European Journal of Cancer*, 163:67–75, 2022. 543 544 545 546
- [33] Yaodong Yu, Stephen Bates, Yi Ma, and Michael I. Jordan. Robust calibration with multi-domain temperature scaling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 547 548 549
- [34] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002. 550 551